



Customer-360 Data Lakehouse & Data Catalog

An event-driven data lakehouse on AWS that unifies 60+ sources into a single, governed 360° view of every customer — powering real-time segmentation, analytics, and multi-channel activation.

INDUSTRY

SCOPE

SOURCES UNIFIED

PLATFORM

Media & Publishing

Data platform ·
Lakehouse

60+

AWS · event-driven

Confidentiality. Sanitized portfolio case study. The client identity, internal project/component names, proprietary business rules, and internal source are withheld under NDA. Screenshots are from a demo/dev environment with account identifiers, personal names, and internal codenames removed; outcome figures are representative and anonymized.

© 2026 Rohail K. Malhi — All rights reserved. Proprietary work; **not open-source**. The frameworks, architecture, metrics, and artifacts shown here may not be copied, adapted, or repurposed. Shared solely to evaluate the author's experience.

AT A GLANCE

CHALLENGE

A media organization's data was siloed across 60+ sources — no unified customer view, no lineage or governance, and standing up a new pipeline took weeks. Real-time, targeted engagement was impossible.

SOLUTION

An event-driven medallion lakehouse on AWS (Raw→Bronze→Silver→Gold) that builds a governed 360° entity view, ships new pipelines in days via a reusable framework, and adds a self-service data catalog.

RESULT

60+ sources unified into one 360° view powering real-time segmentation and multi-channel activation; new pipelines in days; full lineage; and one catalog for all staff to discover and govern data.

60+ → 1

Sources unified into a single 360° entity view

Raw→Gold

Governed medallion lakehouse on S3

Days, not weeks

New pipelines via a reusable, inheritable framework

Self-service

One catalog to discover & govern all data products

CHALLENGE

Siloed data, no single customer view

A large media & publishing organization — dozens of print/digital titles, apps, classifieds, radio, and video — wanted a data-driven growth strategy, but its data was hopelessly fragmented:

- **60+ disconnected sources.** Customer and event data across CRM, ecommerce, web, mobile, ad platforms, video/streaming, social, and partners — no unified collection or activation.
- **No single customer view.** No cross-domain 360° picture of an entity, so real-time targeted engagement was impossible.
- **Slow to add data.** Each new pipeline was a bespoke, weeks-long effort — no reusable framework.
- **No lineage or governance.** Teams couldn't trace data to source or discover existing data products.
- **Tool sprawl.** Every department used different engines/BI tools, with no single place to find and trust data.

SOLUTION

An event-driven medallion lakehouse

A scalable lakehouse on AWS that unifies ingestion, refines data through a governed medallion pipeline into a 360° entity view, orchestrates every job through a decoupled event-driven engine, and exposes it all through a self-service catalog.

The delivery approach

- **Unified ingestion (60+ sources):** one framework to collect, ingest & activate event data — streaming (Kinesis/MSK) and batch (AppFlow, DataSync, SFTP, Glue).
- **Medallion lakehouse:** Raw (as-is) → Bronze (parsed Parquet) → Silver (conformed/typed, Hudi) → Gold (the Customer 360 unified, cross-domain view) — Glue ETL/crawlers, Athena, Lake Formation.
- **Event-driven orchestration:** jobs decoupled from orchestration on an EventBridge bus with ETX/STX/DTX triggers, coordinated by Step Functions/Lambda — no cascading failures.
- **Pipeline framework:** job inheritance + reusable core functionality ships new pipelines in days; jobs deploy to orchestration via GitHub Actions; full lineage/metadata in a graph store (Neptune).
- **Catalog & activation:** a DataHub self-service catalog for all staff; segmentation and multi-channel activation (marketing, loyalty, Email/SMS) plus BI (Athena/QuickSight) and ML (SageMaker).

Architecture at a glance

```
60+ Sources (streaming + batch)
| API Gateway · Kinesis/MSK · AppFlow · DataSync · Glue
S3 Medallion Lakehouse: Raw → Bronze (Parquet) → Silver (Hudi) → Gold · Customer 360
| Glue · Athena · Lake Formation · Glue Catalog
Event-driven Orchestration: EventBridge (ETX/STX/DTX) → SQS → Lambda / Step Functions
├─ lineage: Neptune (graph)      └─ job deploy: GitHub Actions (Terraform)
Consumption & Activation: Athena/Spectrum · QuickSight · SageMaker → segmentation → marketing/loyalty/Email/SMS
Catalog: DataHub (EKS) · Governance: Lake Formation · IaC: Terraform · Helm · Docker · EKS
```

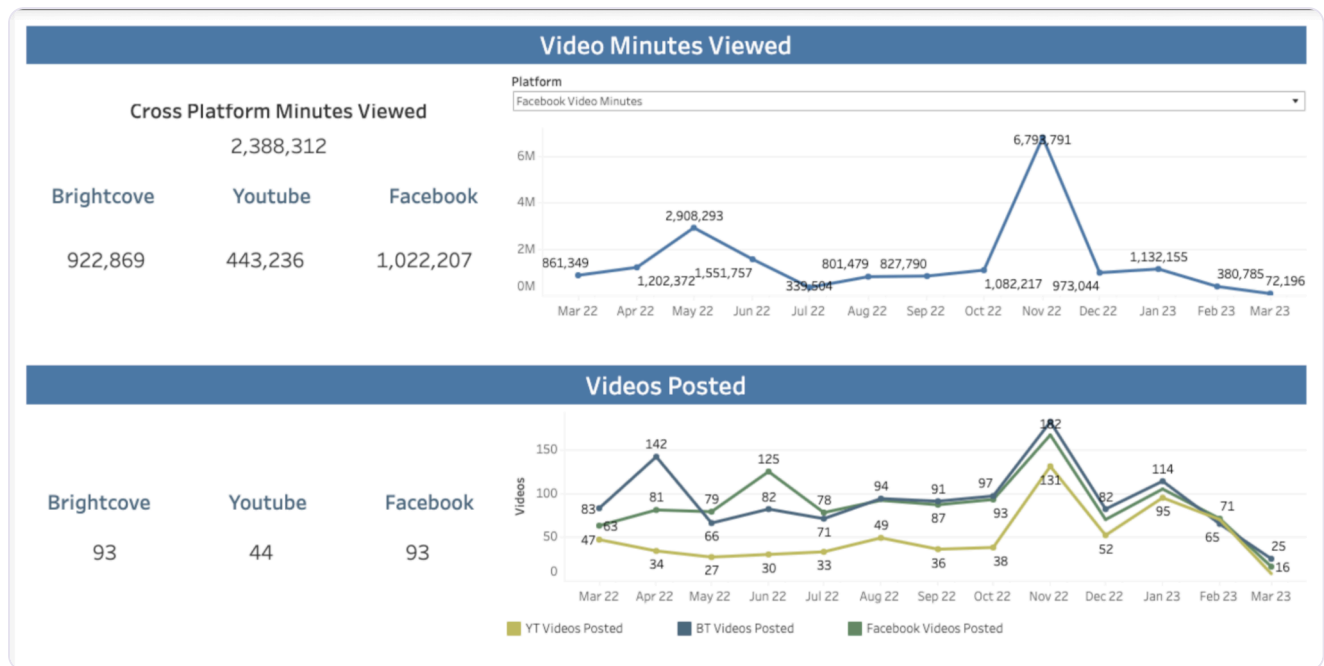
Technology



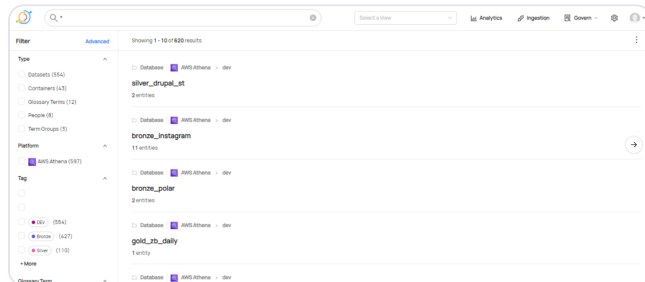
THE PRODUCT

Analytics, catalog & lineage

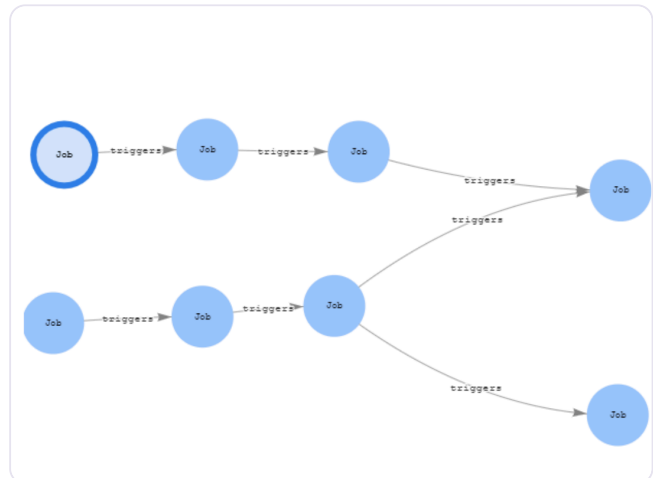
Demo/dev environment. Account IDs, personal names, and internal codenames redacted.



Gold-layer analytics — cross-platform audience metrics (BI on the unified 360° view).



DataHub self-service catalog — hundreds of medallion datasets (bronze/silver/gold), searchable and governed.



Job lineage — decoupled jobs and their trigger dependencies, tracked in a graph store.

RESULT**One governed 360° view — activated****One 360° view**

60+ fragmented sources unified for real-time segmentation

Pipelines in days

Reusable framework + GitHub
Actions job deployment

Traceable

Full lineage & metadata for every schedule and job

- **A single source of truth.** A governed Gold-layer Customer 360 view turned 60+ siloed sources into one cross-domain picture of each customer.
- **Real-time activation.** Audience segmentation off the Gold layer feeds multi-channel activation — marketing, retention, loyalty, personalization, Email/SMS, and social.
- **Faster data delivery.** A standardized, event-driven pipeline framework cut new-pipeline delivery from weeks to days, with consistent logging, tracing, and lineage.
- **A self-service data culture.** A DataHub catalog gave the whole organization one place to discover, understand, and govern data products.
- **Reproducible by design.** The entire platform is codified (Terraform/Helm/Docker/EKS) and replicable across accounts.

Note on figures. Screenshot values are demo/sample data. Specific client data, dataset contents, account identifiers, personal names, and identifying details are excluded under NDA; outcomes here are described qualitatively and anonymized.